SMART PARKING LOT SYSTEM

OVERVIEW

This C++ console application simulates a smart parking lot management system with real-time slot allocation, entry and exit tracking, and dynamic fee calculation.

Team members:

1. Name: Ragesh E
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   Contribution: Core Logic & Entry System (Handles the main structure of the program and vehicle entry process. Implements classes, slot allocation logic, and entry time recording.)
2. Name: Lokesh D
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   Contribution: Exit System & Fee Calculation (Responsible for vehicle exit handling and fee computation. Implements time difference calculation, billing logic, and slot freeing.)
3. Name: Hariharaan Mythili Balasubramanian
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   Contribution: Display, UI & Documentation (Manages menu interface, status display, and available slots view. Prepares test report, sample outputs, and technical README documentation.)

KEY FEATURES

- Dynamic Slot Allocation (First-available algorithm)
- Real-time Entry Time Recording using <ctime>
- Fee Calculation: $2 for the first hour + $1 per additional hour
- Real-time Status Monitoring and Occupancy Statistics
- Vehicle Entry and Exit Management
- Available Slot Display

PROGRAM FLOW

1. Initialize ParkingLot with a capacity of 10 slots
2. Main Menu Loop:
    - Park Vehicle → Find available slot → Record entry time
    - Exit Vehicle → Calculate duration → Compute fee → Free slot
    - Display Status → Show all slots with occupancy details
    - Show Available Slots

CLASS STRUCTURE

ParkingSlot Class

Properties:

- vehicleNumber
- entryTime
- occupied status

Methods:

- parkVehicle()
- exitVehicle()
- calculateFee()

ParkingLot Class

Properties:

- vector<ParkingSlot>
- capacity

Methods:

- parkVehicle()
- exitVehicle()
- displayStatus()

FEE CALCULATION ALGORITHM

fee = 2.0 + max(0, hours - 1) * 1.0

Where:

hours = difftime(currentTime, entryTime) / 3600.0

SAMPLE EXECUTION

Refer to:
03_Test_Report/Example_Execution.txt

COMPILATION

g++ -o parkinglot main.cpp ParkingLot.cpp

RUN

./parkinglot

SAMPLE RUN:

To run:

```
PS C:\Users\rages\OneDrive\Desktop\OOPs case study> g++ -o parkinglot 01_Source_Code/*.cpp
PS C:\Users\rages\OneDrive\Desktop\OOPs case study> ./parkinglot
```

Vehicle Entry (Parking):

```
========================================
       SMART PARKING LOT SYSTEM
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--- MAIN MENU ---
1. Park Vehicle (Entry)
2. Exit Vehicle (Calculate Fee)
3. Display Parking Status
4. Display Available Slots
5. Exit Program
Enter your choice (1-5): 1
Enter vehicle number: 123

Vehicle 123 parked successfully at Slot 1!
Entry Time: Thu Apr  9 16:56:30 2026
```

The user selects option 1 and enters a vehicle number to park.
The system assigns an available slot and records the entry time for tracking.

Parking Status Display:

```
--- MAIN MENU ---
1. Park Vehicle (Entry)
2. Exit Vehicle (Calculate Fee)
3. Display Parking Status
4. Display Available Slots
5. Exit Program
Enter your choice (1-5): 3

PARKING LOT STATUS (10 slots):
Slot    Status      Vehicle #      Entry Time
-----------------------------------------------------
1       OCCUPIED    123            Thu Apr  9 16:56:30 2026
2       AVAILABLE   -              -
3       AVAILABLE   -              -
4       AVAILABLE   -              -
5       AVAILABLE   -              -
6       AVAILABLE   -              -
7       AVAILABLE   -              -
8       AVAILABLE   -              -
9       AVAILABLE   -              -
10      AVAILABLE   -              -

Occupancy: 1/10 (10%)
```

The user selects option 3 to view the current status of all parking slots.
The system shows occupied and available slots along with vehicle number, entry time, and overall occupancy.

Available Slots Display:

```
--- MAIN MENU ---
1. Park Vehicle (Entry)
2. Exit Vehicle (Calculate Fee)
3. Display Parking Status
4. Display Available Slots
5. Exit Program
Enter your choice (1-5): 4

AVAILABLE SLOTS:
Slot 2
Slot 3
Slot 4
Slot 5
Slot 6
Slot 7
Slot 8
Slot 9
Slot 10
```

The user selects option 4 to view only the free parking slots.
The system lists all available slots, helping users quickly find where they can park

Vehicle Exit and Fee Calculation:

```
--- MAIN MENU ---
1. Park Vehicle (Entry)
2. Exit Vehicle (Calculate Fee)
3. Display Parking Status
4. Display Available Slots
5. Exit Program
Enter your choice (1-5): 2
Enter vehicle number to exit: 123

Vehicle Exit Details:
Vehicle: 123
Slot: 1
Entry Time: Thu Apr  9 16:56:30 2026
Total Fee: $2.00
Please pay $2.00
Vehicle exited successfully!
```

The user selects option 2 and enters the vehicle number to exit.
The system calculates parking duration, displays the total fee, and frees the slot after exit.

Program Exit:

```
--- MAIN MENU ---
1. Park Vehicle (Entry)
2. Exit Vehicle (Calculate Fee)
3. Display Parking Status
4. Display Available Slots
5. Exit Program
Enter your choice (1-5): 5
Thank you for using Smart Parking Lot!
```

The user selects option 5 to terminate the application.
The system ends execution and displays a closing message to the user.

Conclusion

The **Smart Parking Lot Management System** successfully demonstrates a simple and efficient way to manage parking operations using C++. It provides essential features such as vehicle entry and exit handling, real-time slot allocation, time tracking, and automatic fee calculation, all through an interactive console-based interface.

The system ensures proper organization of parking slots, reduces manual effort, and improves accuracy in tracking vehicle records. Overall, this project showcases the practical implementation of programming concepts like OOP, file handling (if used), and real-time data processing in solving a real-world problem.